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PAPER_990: F_U_Bi vs F_U_Bi_i Distinction

Abstract

We formalize the distinction between two complementary UQFF buoyancy quantities that are frequently confused:

Property	$F_{U,Bi}$	F_{U,Bi_i}
Direction	Inside $\rightarrow$ Outside	Outside $\rightarrow$ Inside (net)
Sign	Always positive	Typically negative
Units	Energy (J)	Acceleration (m/s ²)
Magnitude	$\sim 10^{40}$	$\sim 10^{-2}$
Physical meaning	Vacuum mass transport	Net gravitational acceleration

1. F_U_Bi (Inside-to-Outside)

$$F_{U,Bi} = \rho_{SCm} \cdot V \cdot S_{26}^2 \cdot \frac{|U_b|}{|U_g| + |U_b|}$$

This is the total vacuum energy flowing outward through the 26-layer buoyancy structure. It is always positive and cosmologically large because it includes $V_{\text{region}} \sim 10^{48} \text{ m}^3$.

2. F_U_Bi_i (Outside-to-Inside)

$$F_{U,Bi_i} = U_g + U_m + U_A - U_b + F_n \cdot S_{26} \cdot \Phi \cdot E_{\text{net}}$$

This is the per-particle net acceleration through the 6-layer canonical structure. Negative values indicate net inward pull (gravity dominates buoyancy at the acceleration level).

3. Complementarity

The ratio $|F_{U,Bi_i}|/F_{U,Bi}$ gives the fractional acceleration per unit vacuum energy — the “efficiency” of gravitational focusing through the buoyancy medium.

4. Implementation

File: fubi_inside_outside.py, class FUBiDistinctionCalc. CP4 class #574.